

Examining Police Surveillance and Crime Rate Correlations in Downtown Atlanta, GA

Emma McMullan

Department of Physics and Astronomy



Introduction

Since its conception in 2007, the Atlanta Police Department's city surveillance system, Connect Atlanta / Operation Shield, has worked to integrate thousands of new cameras into their network annually [1]. With more than 60,000 cameras spanning public and private holders, Atlanta has emerged as the single most-surveilled city in the United States [3]. While camera installment, maintenance, and video storage remains one of the most sizable portions of APD's budget after personnel costs, few comprehensive reviews of the efficacy of this surveillance apparatus in crime prevention exist. Using the locations of APD operated cameras in the Downtown Area (one of the most densely surveilled regions of Atlanta) [2], this project compares reported crime densities from 2023 in regions within camera view and outside of camera view. From this, the extent to which visible surveillance equipment acts as a deterrence to both violent and nonviolent crimes will be compared, examining the value of current surveillance methods as a means of crime prevention in Atlanta.

Methods

APD-operated camera data was obtained from an open records request from the Atlanta Community Press Collective in 2023 [5]. Visible camera locations were then identified manually from their APD labels using Google Maps/Google Street View. License plate readers (LPRs), helicopter monitoring cameras, and cameras that could not be placed through this method were excluded from the mapped dataset. APD cameras were given two designations: pan, tilt, and zoom (PTZ), and quad. Figure 1 shows the locations of these cameras by device type. The Atlanta Police Foundation reports that in 2023, these cameras were primarily manufactured by Axis [4] and had average "seeing" distances of 200 m and 20 m, respectively. Using this, buffer zones were created corresponding to camera type, characterizing surveilled regions. These two zones were then combined into a single shapefile through a union, and then clipped with a shapefile of the Atlanta Downtown neighborhood (as defined by the Atlanta Department of City Planning [6]) to identify the regions of the Downtown area visible to APD camera (shown in Figure 2). A shapefile for the un-surveilled region was also created by erasing the surveilled regions shapefile from the Downtown neighborhood shapefile, allowing these two areas to be compared more directly.

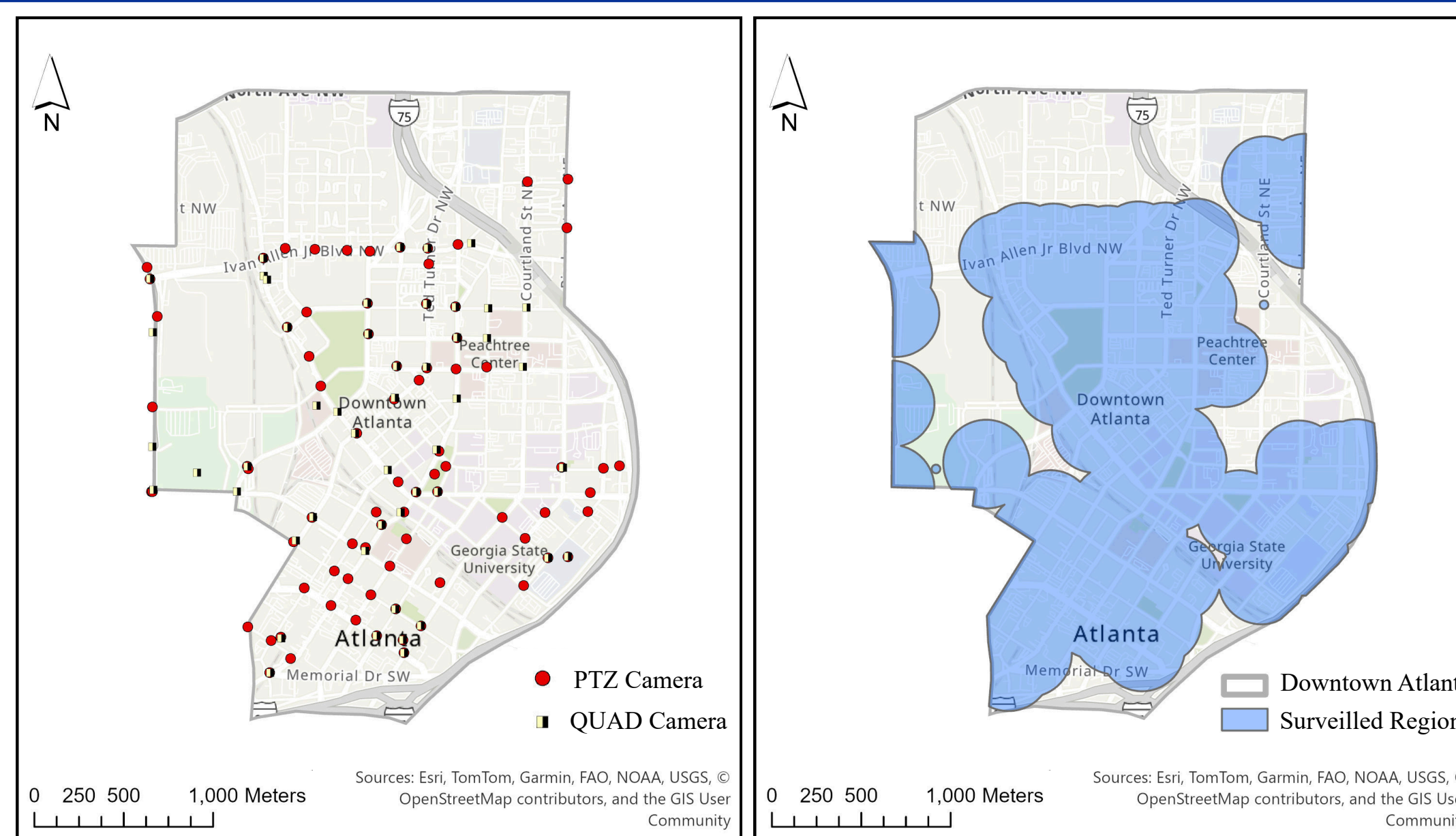


Figure 1: APD Owned Cameras Located in Downtown Atlanta in 2023 by Device Type

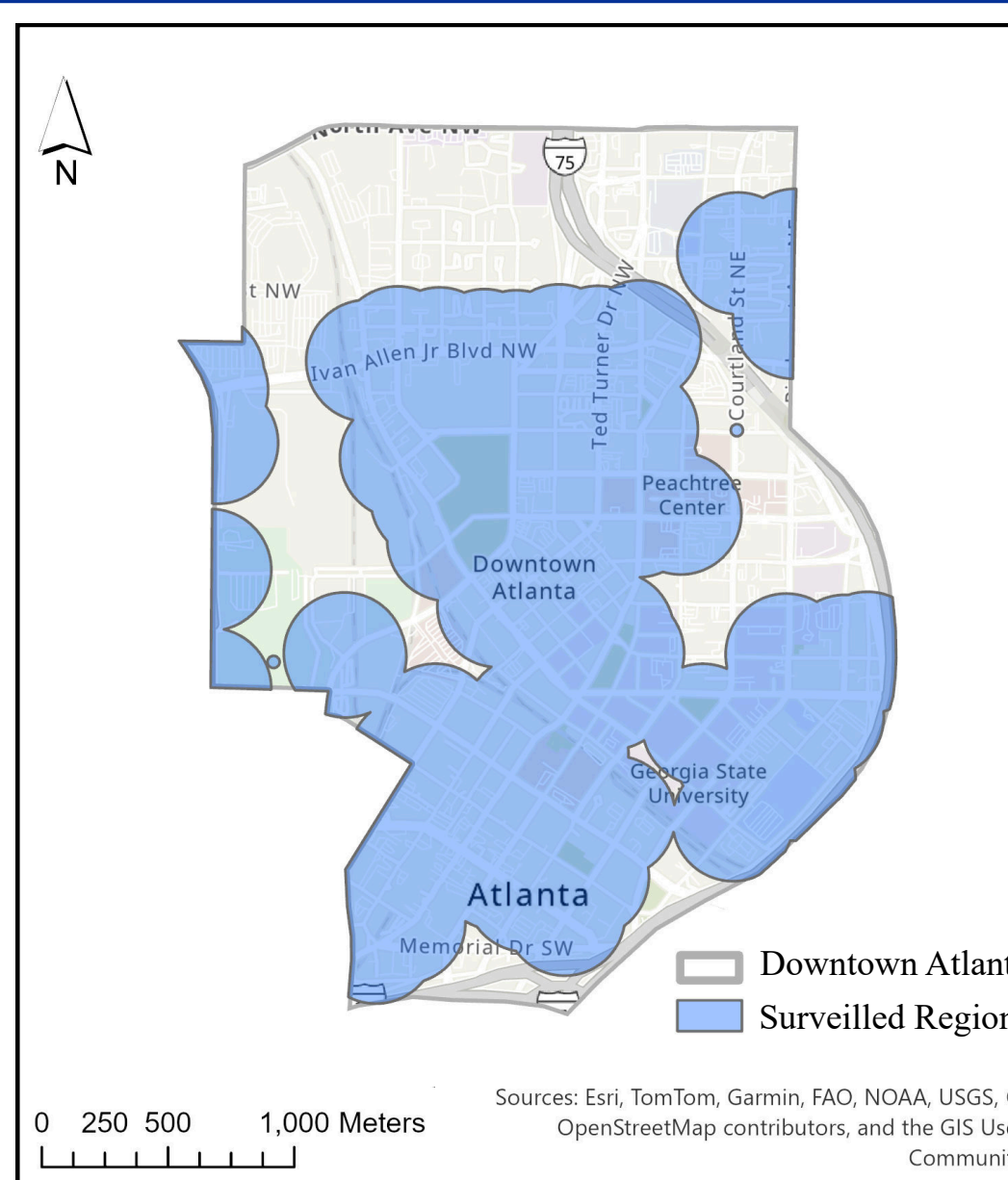


Figure 2: Regions Within APD Operated Camera Purview in Downtown Atlanta in 2023

Crime data for 2023 in the Downtown neighborhood was accessed via the Atlanta Police Open Data Portal [7], where shapefiles were then created for the following crime attributes through the select by attribute tool: crimes against persons, crimes against property, crimes against society, and crimes with or without firearm involvement, as mapped in Figures 3, 4, and 5. These shapefiles were spatially joined with the surveilled and un-surveilled regions, and field geometry was used to calculate each polygon's area. The summary statistics tool was then used to calculate crime densities by type and region.

Results & Discussion

A total of 132 APD-operated cameras were found in Downtown Atlanta in 2023. Cameras were concentrated in the southernmost regions of Downtown, with the majority (127 cameras) located along or south of Ivan Allen Jr. Boulevard. Out of the 5,084,760 m² (~1,257 acres) that comprise the Downtown neighborhood, 65.2% was found to be observable by the APD surveillance network in 2023.

	Surveilled (per 1,000 m ²)	Un-surveilled (per 1,000 m ²)
Crimes Against Persons	0.0866	0.0401
Crimes Against Property	0.441	0.206
Crimes Against Society	0.312	0.1134

In surveilled regions, crime densities were found to be 2-3x higher than in un-surveilled regions for all crime categories, suggesting that the presence of visible security cameras in a given region may not contribute significantly to crime prevention and deterrence.

A total of 358 crimes against persons (such as violent crimes like aggravated assault, intimidation, rape, and fondling) were reported in Downtown in 2023. Of these:

- 71 occurred in un-surveilled areas, 12 of which involved firearms
- 287 occurred in surveilled areas, 52 of which involved firearms

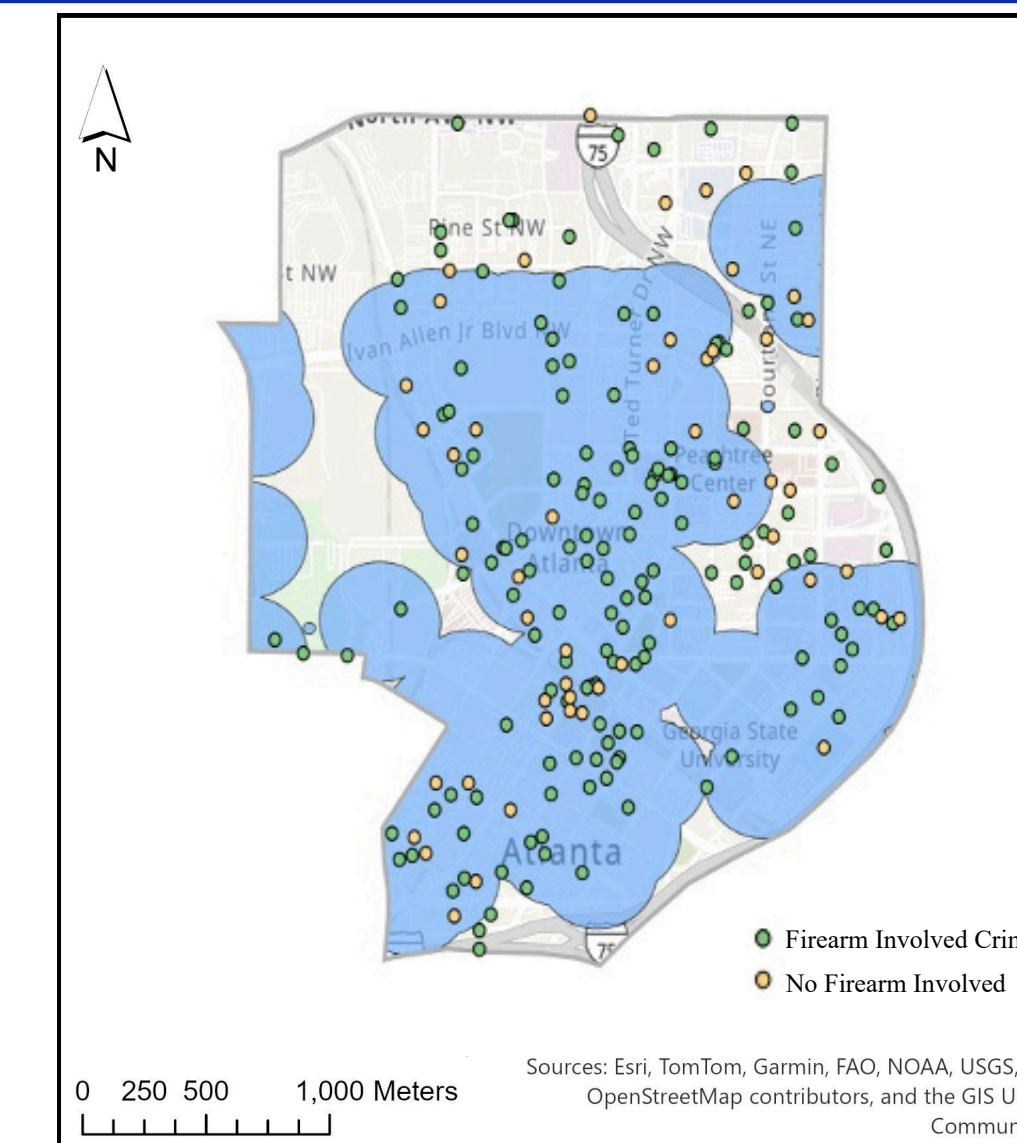


Figure 5: Crimes Against Persons in Downtown Atlanta in 2023, With and Without Firearm Involvement

There were 1,826 crimes against property (including motor vehicle theft, larceny, destruction of property, vandalism) in 2023, where:

- 365 occurred in un-surveilled areas, with 3 incidents involving firearms
- 1,461 occurred in surveilled areas, with 14 involving firearms

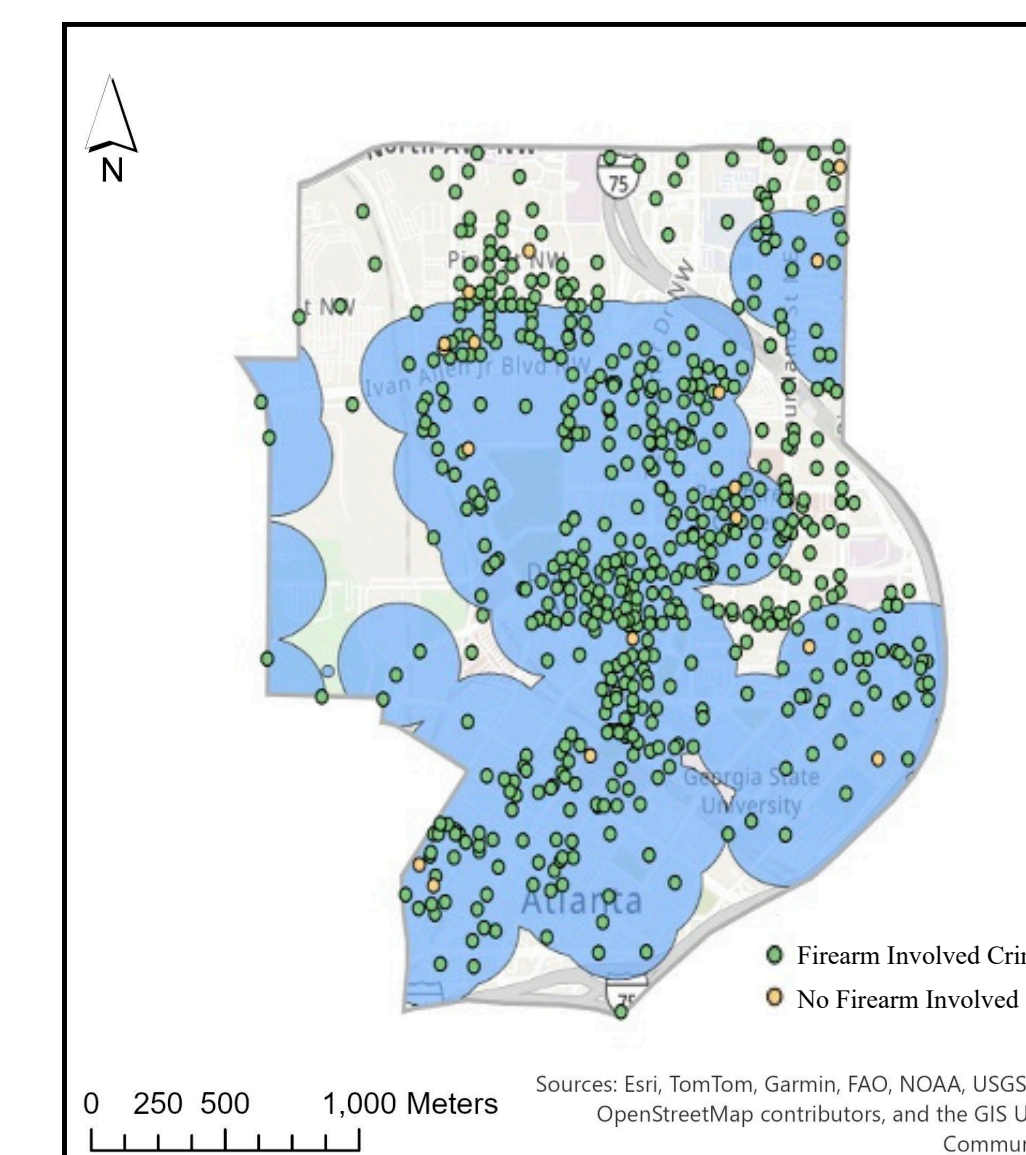


Figure 3: Crimes Against Property in Downtown Atlanta in 2023, With and Without Firearm Involvement

Crimes against society (such as disorderly conduct, narcotics violations, weapons law offenses, and related nonviolent offenses) totaled 1,234 incidents in 2023:

- 201 occurred in un-surveilled areas, with 0 involving firearms
- 1,033 occurred in surveilled areas, with 19 involving firearms

Figures 4, 5, and 6 highlight that across all crime categories, surveilled regions may be associated with higher arrest rates and reported crime densities compared to regions with lower surveillance. The increase in crime densities observed across all mapped categories may suggest that surveillance apparatuses may not be a significant factor in crime prevention, though, mapping could be reflecting camera placement in crime "hotspots," or the contributions of cameras in assisting arrests. Though currently not publicly available, more detailed APD data indicating the extent to which cameras assist individual arrests could clarify the extent to which surveillance contributes to arrest outcomes, allowing for the value of surveillance as a crime deterrent to be better quantified.

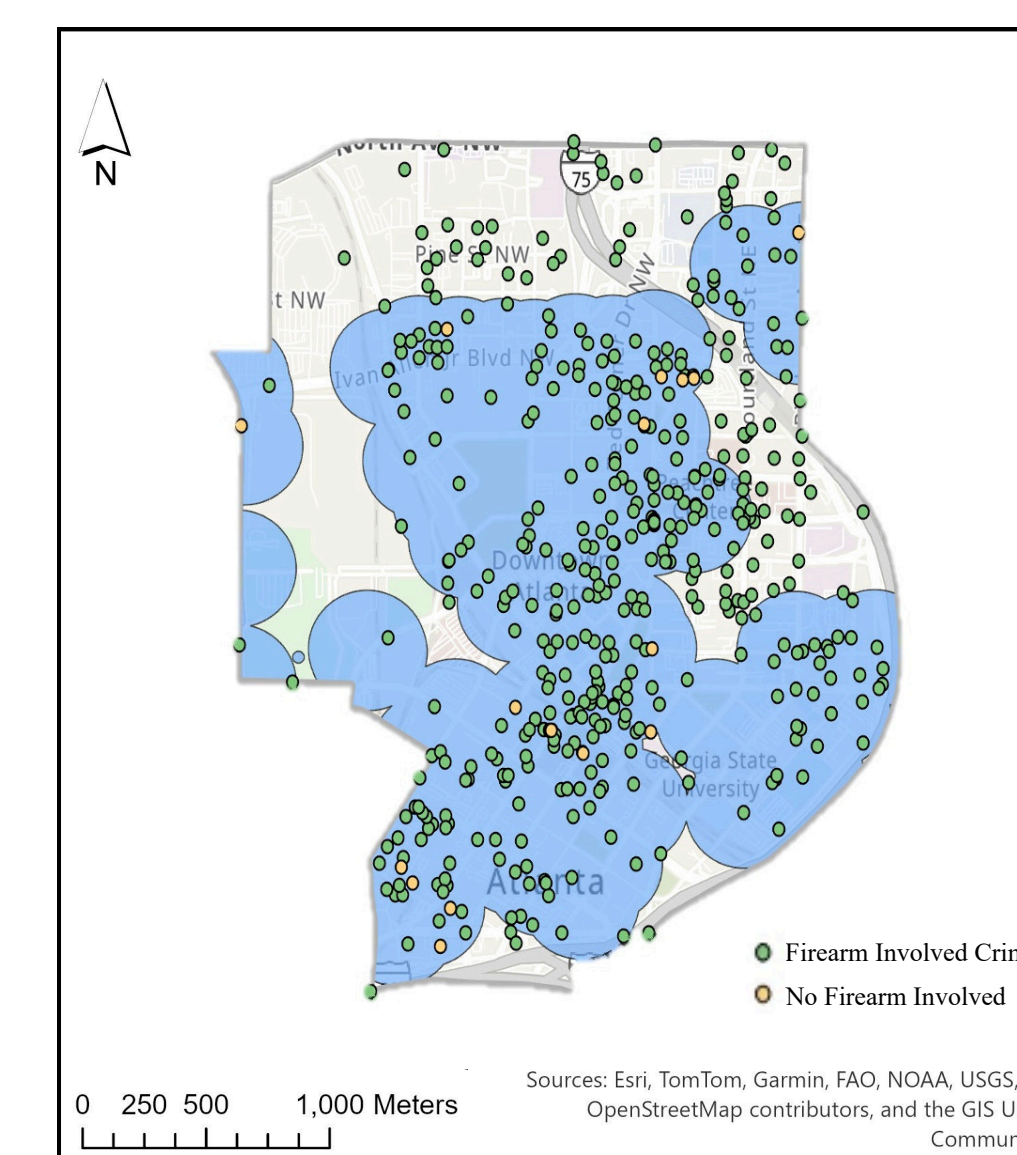


Figure 4: Crimes Against Society in Downtown Atlanta in 2023, With and Without Firearm Involvement

References

- [1] Atlanta Police Foundation. (2025). Operation Shield. <https://atlantapolicefoundation.org/programs/operation-shield/>. [2] Shelton, T. (2025, June 10). Atlanta surveillance camera network mapping with Taylor Shelton. Atlanta Community Press Collective. <https://atpresscollective.com/2025/06/10/atlanta-surveillance-camera-network/>. [3] American Civil Liberties Union-GA. (2025). Speech and privacy. <https://www.acluga.org/issues/speech-and-privacy/>. [4] Atlanta Police Foundation. (2022). Operation Shield: Cameras. <https://sites.google.com/atlantapolicefoundation.org/operation-shield/cameras?authuser=0> [5] Atlanta Community Press Collective. (2025, June 10). APD CAM and LPR list export 202306202 [Open records request]. DocumentCloud. <https://www.documentcloud.org/documents/25969945-apd-cam-and-lpr-list-export-202306202/>. [6] City of Atlanta, Department of City Planning. (2024, August 8). Official Neighborhoods [GIS data set]. ArcGIS Open Data. <https://dpcd-coaplangis.opendata.arcgis.com/datasets/official-neighborhoods-open-data>. [7] Atlanta Police Department. (2025, May 17). Atlanta Police Open Data [Open Data Portal]. <https://opendata.atlantapd.org/>